Sam Rivera

Staff Backend Engineer

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Staff engineer with 12+ years building data-intensive backend systems and developer platforms. Comfortable leading cross-team technical initiatives, mentoring engineers and turning ambiguous problems into reliable, well-documented services. All data in this profile is fictional.

EXPERIENCE

Staff Software Engineer, Tidewater Labs — Jan 2022 - Present
Observability and developer platform company · Lisbon, Portugal
Technical lead of the ingestion platform, a 9-engineer group spanning three teams.

- Redesigned the event ingestion pipeline, cutting p99 latency from 1.8s to 350ms while tripling throughput.
- Introduced service-level objectives and error budgets across 14 services, reducing paging incidents by 45%.
- Mentored six engineers, two of whom were promoted to senior; ran the internal technical-writing guild.

Senior Software Engineer, Lumen Analytics — Sep 2018 - Dec 2021
Business intelligence SaaS · Porto, Portugal
Owned the data-modeling and query services behind customer dashboards.

- Built a columnar query layer that made dashboards 8x faster for the largest customers.
- Led the migration from a single PostgreSQL cluster to a sharded architecture with zero planned downtime.
- Created the internal data-contract process adopted by all product teams.

Full-Stack Developer, Brightpath Software — Jun 2016 - Aug 2018
Product consultancy · Lisbon, Portugal
Delivered web products for clients in logistics, health and education.

- Shipped 11 customer projects end to end, from discovery to production support.
- Standardized the team's CI/CD templates, halving the time to set up a new project.

Software Developer, Cobalt Freight Systems — Jul 2014 - May 2016
Logistics software · Lisbon, Portugal
Developed route-planning and tracking services.

- Implemented a route-optimization service that reduced fleet mileage by 9%.
- Replaced nightly batch jobs with an event-driven tracking pipeline.

PROJECTS

Pipeline Observatory — Open source · Creator, Maintainer — Mar 2021 - Present
Open-source tool that traces data lineage across batch and streaming pipelines.

- 2.4k GitHub stars; used in production by 30+ companies.
- Plugin system with 12 community-contributed connectors.

Go, Kafka, OpenTelemetry

Schema Sentinel — Author — May 2023 - Present
Data-contract linter that blocks breaking schema changes in CI.

- Adopted by 40 repositories at Tidewater Labs within six months.

TypeScript, JSON Schema, GitHub Actions

cv-factory — Oct 2026 - Present
CLI that renders tailored, ATS-friendly CVs from a single structured source of truth.

- Variants, bilingual content and swappable templates from one JSON file.

TypeScript, Typst, JSON Schema

SKILLS

Languages (Expert): TypeScript, SQL
Systems & data languages (Expert): Go, Python
Backend & Data (Expert): Node.js, PostgreSQL, Kafka, Redis, gRPC
Platform & Cloud (Advanced): Kubernetes, Terraform, AWS, Observability

CERTIFICATIONS

Cloud Architecture Professional, Example Cloud Institute — Mar 2025
Kubernetes Operations Specialist, Open Orchestration Foundation — Sep 2024
Secure Software Engineering, Secure Code Alliance — Feb 2024
Data Engineering Professional, Data Guild — Oct 2023
Site Reliability Practitioner, Reliability Institute — Jun 2023
Distributed Systems Fundamentals, Systems School — Aug 2022

EDUCATION

Riverside Institute of Technology, Master of Science, Distributed Computing — 2010 - 2012
GPA 18/20
Courses: Distributed Systems, Database Internals, Performance Engineering

Lakeview University, Bachelor of Science, Computer Science — 2007 - 2010
GPA 16/20

PUBLICATIONS

Error Budgets in Practice, Reliability Engineering Journal — Apr 2024
A field study of adopting service-level objectives across 14 services.

Data Contracts Without the Ceremony, Samrivera.example.com/blog — Aug 2023
A lightweight approach to schema governance for product teams.

LANGUAGES

English — Full professional proficiency
Portuguese — Native
Spanish — Professional working proficiency

INTERESTS

Technical writing: Blogging, Documentation
Trail running